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Data Pipeline Architecture & Design — Tech Blogs by Top Product Companies

A curated collection of the **best tech blog posts** on data pipeline architecture and data pipeline design from **25 top product-based companies**. These are first-party engineering blogs written by the teams who built these systems at scale — ideal for mastering modern data engineering.

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1. Netflix

Netflix processes over **1 trillion events/day** and ingests **~1.3 PB/day**. Their Keystone pipeline processes up to 2 trillion messages per day.

- Evolution of the Netflix Data Pipeline — The foundational blog on how Netflix evolved their data pipeline from batch to real-time using Kafka, S3, and EMR.
- Data Mesh — How Netflix Moves and Processes Data — Netflix's internal "Data Mesh" pipeline for extracting, enriching, and syncing data across systems using Kafka and Flink.
- Real-Time Distributed Graph: Part 1 — Ingesting and Processing Data Streams — Stream processing architecture using Apache Flink to build a real-time distributed graph.
- Real-Time Distributed Graph: Part 2 — Scalable Storage Layer — Storage design for billions of nodes/edges with single-digit ms latencies using Cassandra.
- Behind the Streams: Building a Reliable Cloud Live Streaming Pipeline — Dual-pipeline architecture for live streaming with redundancy and failover.

- How Netflix Built Their Data Pipeline with Amazon Redshift — Deep dive into Netflix's Kafka-centric data stack with Redshift, Hive, Snowflake, and Metacat.

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2. Uber

Uber's data platform supports **156M active customers** across 10,000 cities. They pioneered Apache Hudi and run one of the largest Lambda/Kappa architectures.

- Designing a Production-Ready Kappa Architecture for Data Stream Processing — How Uber designed Kappa architecture to backfill streaming workloads using a unified codebase.
- Building Scalable Streaming Pipelines for Near Real-Time Features — Apache Flink-based streaming pipelines for demand/supply ML features.
- Setting Uber's Transactional Data Lake in Motion with Incremental ETL Using Apache Hudi — Incremental ETL on the lakehouse using Hudi for 100% data completeness.
- From Archival to Access: Config-Driven Data Pipelines — Config-driven archival/retrieval framework built on Airflow and S3.
- Real-Time Data Infrastructure at Uber (Research Paper) — Comprehensive paper covering Kafka, Flink, Pinot, Presto, and the unified architecture.
- Uber's Airflow: 200K Data Pipelines at Massive Scale — How Uber standardized on Airflow (Piper) to orchestrate 200K+ pipelines across 1,000 teams.

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3. Spotify

Spotify processes **1.4 trillion data points daily** and runs **20,000+ batch data pipelines** owned by 300+ teams on GCP.

- Data Platform Explained Part I — Overview of Spotify's data platform building blocks: event delivery, processing, and management.
- Data Platform Explained Part II — Deep dive into event delivery infrastructure (1T+ events/day), data processing, and data management.

- Big Data Processing at Spotify: The Road to Scio (Part 1) — Evolution from Luigi/Scalding/Spark to Scio (Scala API for Apache Beam) on Google Cloud Dataflow.
- Why We Switched Our Data Orchestration Service — Migration from Luigi/Flo to Flyte for pipeline orchestration.
- How Spotify Optimized the Largest Dataflow Job Ever for Wrapped 2020 — Sort Merge Bucket (SMB) join optimization for massive Dataflow pipelines.
- Spotify Unwrapped: How We Brought You a Decade of Data — Architecture for processing 10 years of user listening data for Wrapped.

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4. LinkedIn

LinkedIn pioneered Apache Kafka, and their data lake holds over an **exabyte** of data with hundreds of thousands of pipelines.

- The Log: What Every Software Engineer Should Know About Real-Time Data's Unifying Abstraction — The legendary blog by Jay Kreps on log-centric data pipeline architecture (must-read classic).
- Streaming Data Pipelines with Brooklin — LinkedIn's distributed streaming data pipeline framework for near-real-time applications.
- From Daily Dashboards to Enterprise-Grade Data Pipelines — How LinkedIn evolved from a simple dashboard to enterprise-grade pipelines.
- Declarative Data Pipelines with Hoptimator — Unified control plane for all data pipelines using Flink SQL.
- An Inside Look at LinkedIn's Data Pipeline Monitoring System — Monitoring, alerting, and auto-remediation for data pipelines.
- Evolving LinkedIn's Analytics Tech Stack — Migration from Teradata to Hadoop/Spark with pipeline optimization.
- Towards Data Quality Management at LinkedIn — Data quality at scale: freshness, completeness, and schema monitoring.
- Incremental Data Capture for Oracle Databases — CDC pipeline evolution from batch SQL to near real-time ingestion.

5. Meta (Facebook)

Meta operates one of the **world's largest data warehouses** (300+ PB) and processes billions of queries per second.

- Composable Data Management at Meta — How Meta's data management evolved into a composable architecture using Velox.
- How Meta Understands Data at Scale — DataSchema: describing 100M+ schemas across 100+ data systems.
- How Meta Discovers Data Flows via Lineage at Scale — Data lineage as a critical tool for privacy and understanding data flows.
- Data Logs: The Latest Evolution in Meta's Access Tools — Architecture for processing user data from Hive using Dataswarm pipeline system.
- Meta's Infrastructure Evolution and the Advent of AI — 21 years of infrastructure evolution including data pipeline convergence with AI/ML.

6. Airbnb

Airbnb created **Apache Airflow** and **Apache Superset**, and their data-informed culture has shaped modern data engineering.

- Data Infrastructure at Airbnb — The foundational blog covering Gold/Silver cluster architecture, Airflow, Presto, Spark, and HDFS.
- How Airbnb Built a Key-Value Store for Petabytes of Data (Mussel) — Evolution from HFileService to Mussel: bulk load pipelines and incremental processing.

7. Lyft

Lyft collects **9 trillion analytical events/month** and runs **750K data pipelines** with 400K Spark jobs.

- Orchestrating Data Pipelines at Lyft: Comparing Flyte and Airflow — In-depth comparison of Flyte vs Airflow for pipeline orchestration.

- Real-Time Spatial-Temporal Forecasting at Lyft — Forecasting architecture using Apache Beam/Flink, Kafka, and DynamoDB.

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8. DoorDash

DoorDash processes **220 TB/day** into their data lake using Flink and powers analytics with Delta Lake + Pinot.

- DoorDash Engineering Blog — Data Platform — Collection of posts covering Flink, Delta Lake, Pinot, and Airflow-based pipelines.

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9. Stripe

Stripe handles hundreds of terabytes of daily data transfers and builds reconciliation-first data pipelines for financial accuracy.

- Stripe Engineering Blog — Posts covering database infrastructure, data pipeline design, and the Data Movement Platform.

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10. Pinterest

Pinterest scaled from MySQL to a Kafka-based pipeline supporting **500 million users**.

- How Pinterest Scaled Its Architecture to Support 500M Users — Evolution from MySQL logging to Kafka → S3 pipeline with MapReduce analytics.

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11. Shopify

Shopify processes **57 PB of data** during BFCM, handling 10.5 trillion database queries with Kafka handling **66M messages/second**.

- Shopify's Data Science & Engineering Foundations — Data pipeline principles including unit testing every pipeline job.

- How We're Solving Data Discovery Challenges at Shopify — Metadata ingestion pipeline and data cataloging architecture.
- Data Science Engineering Blog — Collection of data pipeline and engineering posts.

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12. Slack

Slack's data infrastructure uses Hive, Spark, and Presto with Thrift-defined schemas and Parquet storage.

- Data Wrangling at Slack — How Slack manages ETL pipelines across Hive, Spark, and Presto with schema evolution.

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13. Dropbox

Dropbox built a complete analytics data stack from scratch and manages petabytes of user file metadata.

- Dropbox Engineering Blog — Infrastructure — Posts covering data pipeline infrastructure, storage systems, and search indexing.

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14. Grab

Grab (Southeast Asia's super-app) processes millions of ride and delivery events in real-time.

- Real-Time Data Quality Monitoring at Grab — Data contracts in real-time streaming engines with syntactic and semantic tests.
- Grab Engineering Blog — Data pipeline architecture posts covering Kafka, Flink, and data quality.

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15. Instacart

Instacart built modern search infrastructure and metrics stores using incremental processing.

- How Instacart Built a Modern Search Infrastructure — Search pipeline architecture on Postgres.
- Instacart Engineering Blog — Posts on data pipelines, ML infrastructure, and real-time analytics.

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16. Databricks

Databricks pioneered the **Lakehouse Architecture** and Delta Lake, processing data for thousands of enterprises.

- 5 Steps to Intelligent Data Pipelines with Delta Live Tables — Declarative ETL pipeline design with DLT.
- An AI-First Approach to Data Engineering with Lakeflow — AI-powered transformations in ETL pipelines.
- Databricks Architecture Center — Reference architectures for data pipelines and AI.

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17. Cloudflare

Cloudflare built a complete data platform using Iceberg, their own distributed SQL engine, and Arroyo stream processing.

- Announcing the Cloudflare Data Platform — Pipelines (stream processing), R2 Data Catalog (Iceberg), and R2 SQL for petabyte-scale queries.

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18. Google Cloud

Google Cloud provides reference architectures and battle-tested pipeline designs used by thousands of organizations.

- What Data Pipeline Architecture Should I Use? — Overview of batch ETL, streaming, and data analytics pipeline patterns.

- Building Streaming Data Pipelines on Google Cloud — Pub/Sub → Cloud Run/Dataflow → BigQuery pipeline patterns.
- Designing ETL Architecture for Cloud-Native Data Warehousing — ETL reference architecture using Dataflow, Pub/Sub, and BigQuery.
- Building the Data Engineering Driven Organization — Serverless data pipeline reference architecture.

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19. Twitter / X

Twitter processes hundreds of billions of events daily powering the timeline, ads, and recommendations.

- Twitter Engineering Blog — Posts on real-time data processing, stream processing, and ML pipelines.

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20. Walmart

Walmart operates one of the largest retail data platforms, processing billions of transactions.

- Walmart Global Tech Blog — Posts on data lake architecture, real-time analytics, and pipeline orchestration.

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21. Yelp

Yelp pioneered several data pipeline patterns for review and business data processing.

- Yelp Engineering Blog — Data Pipeline — Posts covering streaming architecture, data pipeline design, and Kafka usage.

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22. Confluent (Apache Kafka)

Confluent (founded by the creators of Apache Kafka at LinkedIn) is the authority on event streaming pipelines.

- Confluent Blog — Data Pipelines — Authoritative posts on Kafka-based data pipeline architecture and event streaming patterns.

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23. Etsy

Etsy built data pipelines to support search, recommendations, and marketplace analytics at scale.

- Etsy Engineering Blog — Posts on data infrastructure, search pipelines, and analytics architecture.

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24. WeWork / Marquez (Open Source)

WeWork created Marquez, an open-source metadata and lineage service for data pipelines.

- Marquez — Open Source Data Lineage — Data pipeline metadata collection and lineage tracking.

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25. Booking.com

Booking.com processes massive volumes of travel data for search, pricing, and recommendations.

- Booking.com Tech Blog — Posts on data pipeline architecture, real-time analytics, and A/B testing infrastructure.

📚 Bonus: Cross-Company Architecture Comparisons

These posts compare data architectures across multiple tech giants — great for understanding common patterns.

- Architecture of Giants: Data Stacks at Facebook, Netflix, Airbnb, and Pinterest — Side-by-side comparison of data stacks across four tech giants.
- They Handle 500B Events Daily — Data Engineering Architectures — Netflix, Uber, Spotify, Meta, and Airbnb architecture comparison.
